Grafana offers several resources, but the three most important are

- *Data sources*—Representing backends you want to use in your dashboards or alerts
- *Dashboards*—A way to organize related visualizations
- *Alerts*—Sending you a notification when a certain thing happens (see chapter 7 for greater detail)

Let us now have a look at Grafana and its top-level resources in action. First, we need to stand up a Grafana instance; we do this with the following Docker Compose file that you launch with the, by now, well-known command `docker-compose -f docker-compose.yaml up` (make sure you first go to `ch06/grafana/`).

Listing 6.1 The Docker Compose configuration for the Grafana instance

```
version: "3.5"
services:
  grafana:
    image: grafana/grafana-oss:9.1.4
    ports:
    - "3000:3000"
    volumes:
    - ./grafana.ini:/etc/grafana/grafana.ini
    - ./ds.yaml:/etc/grafana/provisioning/datasources/datasource.yaml
    depends_on:
    - prometheus

  prometheus:
    image: prom/prometheus:latest
    ports:
    - "9090:9090"
    volumes:
    - ./prom-config.yaml:/etc/prometheus/prometheus.yml
```

Provides a Grafana configuration file that simplifies login

Configures Prometheus as the default data source

The grafana.ini file we reference in the Docker Compose file allows us to avoid logging in. Note that at time of writing, Grafana was in version 9; hence, the concepts and screenshots are referencing this version. There were some fundamental changes from Grafana 8 to 9 (unified alerting—that is, alerts being global, rather than on a per-dashboard basis), so keep this in mind when comparing your onscreen output with the screenshots you see here.

Let's begin with Grafana data sources (https://grafana.com/docs/grafana/latest/datasources/, including Amazon CloudWatch, MySQL, Prometheus, and Jaeger, representing backends you can use for visualizations or alerting. If the core data sources are not enough, there are hundreds of plug-ins (https://grafana.com/grafana/plugins/) available, or you can create your own.

Now go to http://localhost:3000/, and then click the gear icon with the Configuration label in the lower part of the left-hand menu. Select the Data Sources menu item. You should see the preconfigured Prometheus data source and, when you click the Add Data Source button, a list of all the available data sources (including potential data source plug-ins you installed).